

Question 1

$$\begin{aligned} 2x_1 + 0x_2 + x_3 &= 1 \\ 2x_1 + 4x_2 - x_3 &= -2 \\ x_1 - 8x_2 - 3x_3 &= 2 \end{aligned}$$

Augmented Matrix:

$$\left[\begin{array}{ccc|c} 2 & 0 & 1 & 1 \\ 2 & 4 & -1 & -2 \\ 1 & -8 & -3 & 2 \end{array} \right]$$

Step 1: Make leading coefficient in column 1 equal to 1

$$R_3 \leftrightarrow R_1 \Rightarrow \left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 2 & 4 & -1 & -2 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Step 2: Make zeros below pivot

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 0 & 20 & 5 & -6 \\ 0 & 16 & 7 & -3 \end{array} \right]$$

Step 3: Make leading coefficient in column 2 equal to 1

$$R_2 \rightarrow \frac{1}{20}R_2$$

$$\left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 0 & 1 & \frac{1}{4} & -\frac{3}{10} \\ 0 & 16 & 7 & -3 \end{array} \right]$$

Step 4: Make zeros below pivot in column 2

$$R_3 \rightarrow R_3 - 16R_2$$

$$\left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 0 & 1 & \frac{1}{4} & -\frac{3}{10} \\ 0 & 0 & 3 & 13.8 \end{array} \right]$$

Step 5: Make leading coefficient in column 3 equal to 1

$$R_3 \rightarrow \frac{1}{3}R_3$$

$$\left[\begin{array}{ccc|c} 1 & -8 & -3 & 2 \\ 0 & 1 & \frac{1}{4} & -\frac{3}{10} \\ 0 & 0 & 1 & \frac{19}{15} \end{array} \right]$$

Step 6: Back-substitution

$$x_3 = \frac{19}{15}$$

$$x_2 = -\frac{3}{10} - \frac{1}{4} \times \frac{19}{15} = -\frac{89}{60}$$

$$x_1 = 2 + 8 \times \frac{89}{60} + 3 \times \frac{19}{15} = \frac{263}{60}$$

$$\boxed{x_1 = \frac{263}{60}, x_2 = -\frac{89}{60}, x_3 = \frac{19}{15}}$$

Would you like me to continue solving the rest of the questions in this format?

< 2/2 >       

yes but step by step

< 2/2 >

Alright! I'll solve each question step by step, showing all necessary calculations without skipping steps.
Here's **Question 2** using **Gaussian Elimination**:

Question 2

$$\begin{aligned}x_1 + x_2 + x_3 &= a \\x_1 + (1 + a)x_2 + x_3 &= 2a \\x_1 + x_2 + (1 + a)x_3 &= 3a\end{aligned}$$

Step 1: Convert to Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 1 & 1+a & 1 & 2a \\ 1 & 1 & 1+a & 3a \end{array} \right]$$

Step 2: Make first column's leading coefficient 1 (already done)**Step 3: Make zeros below first pivot**

Subtract Row 1 from Row 2 and Row 3:

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & a & 0 & a \\ 0 & 0 & a & 2a \end{array} \right]$$

Step 4: Make leading coefficient in second column 1

Divide Row 2 by a:

$$R_2 \rightarrow \frac{1}{a}R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 0 & 1 \\ 0 & 0 & a & 2a \end{array} \right]$$

Step 5: Make zero above and below second pivot

Since there's no term above the second pivot to eliminate, we move to Row 3. Divide Row 3 by a:

$$R_3 \rightarrow \frac{1}{a}R_3$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Step 6: Back-substitution

$$\begin{aligned}x_3 &= 2 \\x_2 &= 1 \\x_1 &= a - x_2 - x_3 = a - 1 - 2 = a - 3\end{aligned}$$

Final Answer

$$x_1 = a - 3, \quad x_2 = 1, \quad x_3 = 2$$

I'll now solve Question 3 in the same detailed manner. Let me know if you want any modifications in the format!

yes

Question 3

$$\begin{aligned}x_1 - x_2 + x_3 - x_4 + x_5 &= 1 \\2x_1 + x_2 + 3x_3 - 4x_5 &= 3 \\3x_1 - 2x_2 + 2x_3 + x_4 + x_5 &= 1 \\0x_1 + x_2 + 0x_3 + x_4 + x_5 &= 0\end{aligned}$$

Step 1: Convert to Augmented Matrix

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 1 & 1 \\ 2 & 1 & 3 & 0 & -4 & 3 \\ 3 & -2 & 2 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{array} \right]$$

Step 2: Make the first column's leading coefficient 1 (already done)**Step 3: Make zeros below the first pivot**

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & | & 1 \\ 0 & 3 & 1 & 2 & -6 & | & 1 \\ 0 & 1 & -1 & 4 & -2 & | & -2 \\ 0 & 1 & 0 & 1 & 1 & | & 0 \end{bmatrix}$$

Step 4: Swap Row 2 and Row 4 for a simpler second pivot

$$R_2 \leftrightarrow R_4$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & | & 1 \\ 0 & 1 & 0 & 1 & 1 & | & 0 \\ 0 & 3 & 1 & 2 & -6 & | & 1 \\ 0 & 1 & -1 & 4 & -2 & | & -2 \end{bmatrix}$$

Step 5: Make zeros below the second pivot

$$R_3 \rightarrow R_3 - 3R_2, \quad R_4 \rightarrow R_4 - R_2$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & | & 1 \\ 0 & 1 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 1 & -1 & -9 & | & 1 \\ 0 & 0 & -1 & 3 & -3 & | & -2 \end{bmatrix}$$

Step 6: Make the third column's leading coefficient 1 (already done)**Step 7: Make zero below third pivot**

$$R_4 \rightarrow R_4 + R_3$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & | & 1 \\ 0 & 1 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 1 & -1 & -9 & | & 1 \\ 0 & 0 & 0 & 2 & -12 & | & -1 \end{bmatrix}$$

Step 8: Make the fourth column's leading coefficient 1

$$R_4 \rightarrow \frac{1}{2}R_4$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & | & 1 \\ 0 & 1 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 1 & -1 & -9 & | & 1 \\ 0 & 0 & 0 & 1 & -6 & | & -\frac{1}{2} \end{bmatrix}$$

Step 9: Back-substitution

$$x_4 = -\frac{1}{2} + 6x_5$$

$$x_3 = 1 + x_4 + 9x_5 = 1 - \frac{1}{2} + 6x_5 + 9x_5 = \frac{1}{2} + 15x_5$$

$$x_2 = -x_4 - x_5 = -(-\frac{1}{2} + 6x_5) - x_5 = \frac{1}{2} - 6x_5 - x_5 = \frac{1}{2} - 7x_5$$

$$x_1 = 1 + x_2 - x_3 + x_4 - x_5$$

$$= 1 + (\frac{1}{2} - 7x_5) - (\frac{1}{2} + 15x_5) + (-\frac{1}{2} + 6x_5) - x_5$$

$$= 1 + \frac{1}{2} - 7x_5 - \frac{1}{2} - 15x_5 - \frac{1}{2} + 6x_5 - x_5$$

$$= 1 - 17x_5 - \frac{1}{2}$$

$$= \frac{2}{2} - \frac{1}{2} - 17x_5 = \frac{1}{2} - 17x_5$$

Final Answer

$$x_1 = \frac{1}{2} - 17x_5, \quad x_2 = \frac{1}{2} - 7x_5, \quad x_3 = \frac{1}{2} + 15x_5, \quad x_4 = -\frac{1}{2} + 6x_5, \quad x_5 = \text{free variable}$$

Let me know if you want to continue with **Question 4** in the same detailed manner!

yes

Question 4

$$\begin{aligned}
 x_1 + x_2 - x_3 &= 1 \\
 x_2 + x_3 - x_4 &= 1 \\
 x_3 + x_4 - x_5 &= 1 \\
 -x_3 + x_4 + x_5 &= 1 \\
 -x_2 + x_3 + x_4 &= 1
 \end{aligned}$$

Step 1: Convert to Augmented Matrix

$$\left[\begin{array}{ccccc|c}
 1 & 1 & -1 & 0 & 0 & 1 \\
 0 & 1 & 1 & -1 & 0 & 1 \\
 0 & 0 & 1 & 1 & -1 & 1 \\
 0 & 0 & -1 & 1 & 1 & 1 \\
 0 & -1 & 1 & 1 & 0 & 1
 \end{array} \right]$$

Step 2: Make the first pivot 1 (already done)

The first pivot is 1 in the (1,1) position, so we move to making zeros below it.

Since there's already a zero in R2, R3, R4, and R5 under the first column, we move to the next column.

Step 3: Make second pivot 1

The second pivot is 1 in R2, (2,2 position), so it's already in the correct form.

Now, we make zeros below it:

$$\begin{aligned}
 R_5 &\rightarrow R_5 + R_2 \\
 \left[\begin{array}{ccccc|c}
 1 & 1 & -1 & 0 & 0 & 1 \\
 0 & 1 & 1 & -1 & 0 & 1 \\
 0 & 0 & 1 & 1 & -1 & 1 \\
 0 & 0 & -1 & 1 & 1 & 1 \\
 0 & 0 & 2 & 0 & 0 & 2
 \end{array} \right]
 \end{aligned}$$

Step 4: Make third pivot 1

Divide R5 by 2:

$$\begin{aligned}
 R_5 &\rightarrow \frac{1}{2}R_5 \\
 \left[\begin{array}{ccccc|c}
 1 & 1 & -1 & 0 & 0 & 1 \\
 0 & 1 & 1 & -1 & 0 & 1 \\
 0 & 0 & 1 & 1 & -1 & 1 \\
 0 & 0 & -1 & 1 & 1 & 1 \\
 0 & 0 & 1 & 0 & 0 & 1
 \end{array} \right]
 \end{aligned}$$

Now, we make a zero at position (4,3):

$$\begin{aligned}
 R_4 &\rightarrow R_4 + R_5 \\
 \left[\begin{array}{ccccc|c}
 1 & 1 & -1 & 0 & 0 & 1 \\
 0 & 1 & 1 & -1 & 0 & 1 \\
 0 & 0 & 1 & 1 & -1 & 1 \\
 0 & 0 & 0 & 1 & 1 & 2 \\
 0 & 0 & 1 & 0 & 0 & 1
 \end{array} \right]
 \end{aligned}$$

Now, we make a zero at position (3,3) by subtracting R5 from R3:

$$\begin{aligned}
 R_3 &\rightarrow R_3 - R_5 \\
 \left[\begin{array}{ccccc|c}
 1 & 1 & -1 & 0 & 0 & 1 \\
 0 & 1 & 1 & -1 & 0 & 1 \\
 0 & 0 & 0 & 1 & -1 & 0 \\
 0 & 0 & 0 & 1 & 1 & 2 \\
 0 & 0 & 1 & 0 & 0 & 1
 \end{array} \right]
 \end{aligned}$$

Step 5: Make fourth pivot 1 (already done)

Since the pivot at (4,4) is already 1, we proceed to eliminate the (5,4) term:

$$R_5 \rightarrow R_5 - R_4$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 & 0 & | & 1 \\ 0 & 1 & 1 & -1 & 0 & | & 1 \\ 0 & 0 & 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & 1 & 1 & | & 2 \\ 0 & 0 & 1 & -1 & -1 & | & -1 \end{bmatrix}$$

Now, make a zero at (5,3) by subtracting R3 from R5:

$$R_5 \rightarrow R_5 - R_3$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 & 0 & | & 1 \\ 0 & 1 & 1 & -1 & 0 & | & 1 \\ 0 & 0 & 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & 1 & 1 & | & 2 \\ 0 & 0 & 0 & -1 & 0 & | & -1 \end{bmatrix}$$

Now, make R5,4 = 1:

$$R_5 \rightarrow -R_5$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 & 0 & | & 1 \\ 0 & 1 & 1 & -1 & 0 & | & 1 \\ 0 & 0 & 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & 1 & 1 & | & 2 \\ 0 & 0 & 0 & 1 & 0 & | & 1 \end{bmatrix}$$

Step 6: Back-substitution

From Row 5:

$$x_4 = 1$$

From Row 4:

$$x_4 + x_5 = 2 \Rightarrow x_5 = 2 - 1 = 1$$

From Row 3:

$$x_4 - x_5 = 0 \Rightarrow 1 - 1 = 0 \Rightarrow \text{Correct}$$

From Row 2:

$$x_2 + x_3 - x_4 = 1$$

$$x_2 + x_3 - 1 = 1 \Rightarrow x_2 + x_3 = 2$$

From Row 1:

$$x_1 + x_2 - x_3 = 1$$

$$x_1 + x_2 - x_3 = 1$$

Since we have two equations and two unknowns, we solve:

$$x_2 = 2 - x_3$$

$$x_1 + (2 - x_3) - x_3 = 1$$

$$x_1 + 2 - 2x_3 = 1$$

$$x_1 = 1 + 2x_3 - 2$$

$$x_1 = -1 + 2x_3$$

Final Answer

$$x_1 = -1 + 2x_3, \quad x_2 = 2 - x_3, \quad x_3 = \text{free variable}, \quad x_4 = 1, \quad x_5 = 1$$

Let me know if you want me to continue with **Question 5** in the same detailed manner! 😊

yes

Question 5

$$\begin{aligned}x_1 - 2x_2 - 7x_3 + 7x_4 &= 5 \\ -x_1 + 2x_2 + 8x_3 - 5x_4 &= -7 \\ 3x_1 - 4x_2 - 17x_3 + 13x_4 &= 14 \\ 2x_1 - 2x_2 - 11x_3 + 8x_4 &= 7\end{aligned}$$

Step 1: Convert to Augmented Matrix

$$\left[\begin{array}{cccc|c} 1 & -2 & -7 & 7 & 5 \\ -1 & 2 & 8 & -5 & -7 \\ 3 & -4 & -17 & 13 & 14 \\ 2 & -2 & -11 & 8 & 7 \end{array} \right]$$

Step 2: Make the first pivot 1 (already done)

The first pivot is 1 in the (1,1) position, so we move to making zeros below it.

$$R_2 \rightarrow R_2 + R_1, \quad R_3 \rightarrow R_3 - 3R_1, \quad R_4 \rightarrow R_4 - 2R_1$$

$$\left[\begin{array}{cccc|c} 1 & -2 & -7 & 7 & 5 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 2 & 4 & -8 & -1 \\ 0 & 2 & 3 & -6 & -3 \end{array} \right]$$

Step 3: Make the second pivot 1

Since $R_{2,2} = 0$, we swap R_2 and R_3 .

$$R_2 \leftrightarrow R_3$$

$$\left[\begin{array}{cccc|c} 1 & -2 & -7 & 7 & 5 \\ 0 & 2 & 4 & -8 & -1 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 2 & 3 & -6 & -3 \end{array} \right]$$

Now, make zeros below the second pivot:

$$R_4 \rightarrow R_4 - R_2, \quad R_2 \rightarrow \frac{1}{2}R_2$$

$$\left[\begin{array}{cccc|c} 1 & -2 & -7 & 7 & 5 \\ 0 & 1 & 2 & -4 & -\frac{1}{2} \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & -1 & -2 & -\frac{5}{2} \end{array} \right]$$

Step 4: Make the third pivot 1 (already done)

Now, make a zero below the third pivot.

$$R_4 \rightarrow R_4 + R_3$$

$$\left[\begin{array}{cccc|c} 1 & -2 & -7 & 7 & 5 \\ 0 & 1 & 2 & -4 & -\frac{1}{2} \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & -\frac{9}{2} \end{array} \right]$$

Since the last row is $0 = -9/2$, which is a contradiction, this system has **no solution**.

Final Answer

No Solution (Inconsistent System)

Let me know if you want me to continue with **Question 6!** 😊

yes

Question 6

$$\begin{aligned}x_1 + 2x_2 + x_3 &= -1 \\6x_1 + x_2 + x_3 &= -4 \\2x_1 - 3x_2 - x_3 &= 0 \\x_1 - x_2 &= 1\end{aligned}$$

Step 1: Convert to Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 6 & 1 & 1 & -4 \\ 2 & -3 & -1 & 0 \\ 1 & -1 & 0 & 1 \end{array} \right]$$

Step 2: Make the first pivot 1 (already done)

Now, make zeros below it.

$$R_2 \rightarrow R_2 - 6R_1, \quad R_3 \rightarrow R_3 - 2R_1, \quad R_4 \rightarrow R_4 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 0 & -11 & -5 & 2 \\ 0 & -7 & -3 & 2 \\ 0 & -3 & -1 & 2 \end{array} \right]$$

Step 3: Make the second pivot 1

Divide R2 by -11 to make the pivot 1:

$$R_2 \rightarrow \frac{1}{-11}R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 0 & 1 & \frac{5}{11} & -\frac{2}{11} \\ 0 & -7 & -3 & 2 \\ 0 & -3 & -1 & 2 \end{array} \right]$$

Now, make zeros below the second pivot.

$$R_3 \rightarrow R_3 + 7R_2, \quad R_4 \rightarrow R_4 + 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 0 & 1 & \frac{5}{11} & -\frac{2}{11} \\ 0 & 0 & \frac{32}{11} & -\frac{8}{11} \\ 0 & 0 & \frac{4}{11} & \frac{10}{11} \end{array} \right]$$

Step 4: Make the third pivot 1

Multiply R3 by 11/32:

$$R_3 \rightarrow \frac{11}{32}R_3$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 0 & 1 & \frac{5}{11} & -\frac{2}{11} \\ 0 & 0 & 1 & -\frac{8}{32} \\ 0 & 0 & \frac{4}{11} & \frac{10}{11} \end{array} \right]$$

Now, make **zero** below the third pivot.

$$R_4 \rightarrow R_4 - \frac{4}{11}R_3$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & -1 \\ 0 & 1 & \frac{5}{11} & -\frac{2}{11} \\ 0 & 0 & 1 & -\frac{1}{4} \\ 0 & 0 & 0 & \frac{7}{11} \end{array} \right]$$

Since the last row is $0 = 7/11$, which is a contradiction, this system has **no solution**.

Final Answer

No Solution (Inconsistent System)

Let me know if you want me to continue with **Question 7!** 😊

yes

Question 7

$$\begin{aligned} 2x_1 + x_2 + 5x_3 &= 4 \\ 3x_1 - 2x_2 + 2x_3 &= 2 \\ 5x_1 - 8x_2 - 4x_3 &= 1 \end{aligned}$$

Step 1: Convert to Augmented Matrix

$$\left[\begin{array}{ccc|c} 2 & 1 & 5 & 4 \\ 3 & -2 & 2 & 2 \\ 5 & -8 & -4 & 1 \end{array} \right]$$

Step 2: Make the first pivot 1

Divide R1 by 2:

$$R_1 \rightarrow \frac{1}{2}R_1$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{5}{2} & 2 \\ 3 & -2 & 2 & 2 \\ 5 & -8 & -4 & 1 \end{array} \right]$$

Now, make zeros below the first pivot.

$$R_2 \rightarrow R_2 - 3R_1, \quad R_3 \rightarrow R_3 - 5R_1$$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{5}{2} & 2 \\ 0 & -\frac{7}{2} & -\frac{11}{2} & -4 \\ 0 & -\frac{21}{2} & -\frac{29}{2} & -9 \end{array} \right]$$

Step 3: Make the second pivot 1

Multiply R2 by $-2/7$:

$$R_2 \rightarrow \frac{-2}{7}R_2$$

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{5}{2} & 12 \\ 0 & 1 & \frac{11}{7} & 1\frac{8}{7} \\ 0 & -\frac{21}{2} & -\frac{29}{2} & 1-9 \end{bmatrix}$$

Now, make zero below the second pivot.

$$R_3 \rightarrow R_3 + \frac{21}{2}R_2$$

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{5}{2} & 12 \\ 0 & 1 & \frac{11}{7} & 1\frac{8}{7} \\ 0 & 0 & \frac{14}{7} & 13 \end{bmatrix}$$

Step 4: Make the third pivot 1

Multiply R3 by 7/14:

$$R_3 \rightarrow \frac{7}{14}R_3$$

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{5}{2} & 12 \\ 0 & 1 & \frac{11}{7} & 1\frac{8}{7} \\ 0 & 0 & 1 & \frac{13}{2} \end{bmatrix}$$

Now, make zeros above the third pivot.

$$R_1 \rightarrow R_1 - \frac{5}{2}R_3, \quad R_2 \rightarrow R_2 - \frac{11}{7}R_3$$

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{14} \\ 0 & 0 & 1 & \frac{13}{2} \end{bmatrix}$$

Step 5: Back-substitution

From R3:

$$x_3 = \frac{3}{2}$$

From R2:

$$x_2 = \frac{1}{14}$$

From R1:

$$x_1 = \frac{1}{2}$$

Final Answer

$$x_1 = \frac{1}{2}, \quad x_2 = \frac{1}{14}, \quad x_3 = \frac{3}{2}$$

This completes Exercise 4.1 with step-by-step Gaussian elimination! Let me know if you need more practice or explanations. 😊